RIS-DRL 2.0 Draft

RIS-DRL 2.0 Team

I. System Model and Problem Formulation

A. Signal Model

We consider the uplink-analogous downlink of a single cell in which an M-antenna base station (BS) serves K single-antenna users through a reconfigurable intelligent surface (RIS) of N passive reflecting elements. The discrete-time baseband observation at user k is

$$y_k \;=\; \tilde{h}_k(\boldsymbol{\theta})^H w_k \;+\; n_k, \qquad k = 1, \dots, K, \tag{1}$$

where $w_k \in \mathbb{C}^M$ is the BS precoder vector for user k, $n_k \sim \mathcal{CN}(0, \sigma^2)$ is additive white Gaussian noise, and $\tilde{h}_k(\boldsymbol{\theta}) \in \mathbb{C}^M$ is the *effective* BS–user k channel, which concatenates the direct link $h_{d,k} \in \mathbb{C}^M$ with the cascaded BS–RIS–user link through the RIS phase profile $\boldsymbol{\Phi} = \mathrm{diag}(e^{j\theta_1}, \dots, e^{j\theta_n}, \dots, e^{j\theta_N})$:

$$\tilde{h}_k(\boldsymbol{\theta}) \;=\; h_{d,k} \;+\; G^H \boldsymbol{\Phi}^H h_{r,k}. \tag{2}$$

Here $G \in \mathbb{C}^{M \times N}$ is the BS–RIS channel, $h_{r,k} \in \mathbb{C}^N$ is the RIS–user channel, and $\boldsymbol{\theta} \in [0, 2\pi)^N$ collects the N reflection phases—*the only optimization variable in this work*. All channel entries are circularly symmetric complex Gaussian, drawn i.i.d. and scaled by 3GPP urban-micro path-loss models: NLoS for the direct link ($128.1{+}37.6 \log_{10} d_{\mathrm{km}}$ dB) and LoS for both RIS hops ($61.4 + 35.8 \log_{10} d_{\mathrm{km}}$ dB each). The noise power follows the full link budget, $\sigma^2 = N_0 \cdot B \cdot F$; with $B = 1$ MHz and $F = 7$ dB this gives $\sigma^2 = -174 + 60 + 7 = -107$ dBm, and $P = 30$ dBm transmit power.

B. Zero-Forcing Precoding and the Sum-Rate Objective

For any fixed phase vector we apply zero-forcing (ZF) precoding. Stacking the effective channels into $\tilde{H}(\boldsymbol{\theta}) \in \mathbb{C}^{K \times M}$,

$$W \;=\; \tilde{H}^H \big(\tilde{H}\tilde{H}^H\big)^{-1}, \qquad W \leftarrow W \,\frac{\sqrt{P}}{\|W\|_F}, \tag{3}$$

where the power normalization enforces $\sum_k \|w_k\|^2 = P$. ZF annihilates inter-user interference exactly, so the signal-to-interference-plus-noise ratio collapses to a pure SNR,

$$\mathrm{SNR}_k(\boldsymbol{\theta}) \;=\; \frac{\big|\tilde{h}_k(\boldsymbol{\theta})^H w_k\big|^2}{\sigma^2}, \qquad R(\boldsymbol{\theta}) \;=\; \sum_{k=1}^{K} \log_2\!\big(1{+}\mathrm{SNR}_k(\boldsymbol{\theta})\big). \tag{4}$$

This design is deliberately *noise-limited* (cf. Section **??**): because ZF removes interference, SNR_k depends directly on the phase-alignment quality of the cascaded term $h_{r,k}^H \boldsymbol{\Phi} G$, and a misaligned phase profile costs decibels. Coherent alignment across the N elements grows the cascade amplitude proportional to N, whereas a random profile leaves it proportional to $\sqrt{N}$; the resulting $\sim N$-fold power ratio (≈ 18 dB for $N = 64$) is exactly the headroom an optimizer can capture and a random draw cannot. Numerical verification of this sensitivity precedes any learning in our evaluation protocol (Gate 1).

C. Phase Optimization on the Torus

Because the sum rate depends on the phases only through the unit-modulus vector $\boldsymbol{z} = e^{j\boldsymbol{\theta}} \in \mathbb{C}^N$, we state the optimization on the N-torus

$$\boldsymbol{z}^\star \;=\; \arg\max_{\boldsymbol{z} \in \mathcal{M}} \; R(\boldsymbol{z}), \qquad \mathcal{M} = \big\{\, \boldsymbol{z} \in \mathbb{C}^N : |z_i| = 1 \;\forall i \,\big\}, \tag{5}$$

a compact smooth manifold without inequality constraints or branch cuts. The Euclidean gradient of R with respect to the complex vector $\boldsymbol{z}$ is obtained by reverse-mode automatic differentiation through (3)–(4), and is verified against central finite differences to a relative accuracy better than 10^{-5} before any optimization run (Section I-F). All methods studied in this paper—model-free deep reinforcement learning, deep unfolding, and the Riemannian bound—share this single implementation of (4) and are evaluated on one frozen dataset of 500 channels that is generated once and never regenerated, so every comparison is exactly paired.

The default geometry used throughout is $M = 8$, $N = 64$, $K = 4$; the $M = 4$ ablation and the $N = 36$ debug configuration are reported once in Section **??**.

D. Deep Unfolding: A Learned Projected-Gradient Optimizer

The model-free agents of Section **??** must discover from scratch that maximizing the sum rate calls for ascent on (4). Model-driven design instead builds that knowledge in explicitly: we *unfold* L projected-gradient ascent steps on the true objective into a fixed-depth network and train the few remaining degrees of freedom end-to-end. Concretely, the L-layer optimizer executes

$$\boldsymbol{\theta}^{(l+1)} \;=\; \mathrm{wrap}\Big(\boldsymbol{\theta}^{(l)} {+} \alpha_l \,\nabla_{\boldsymbol{\theta}} R\big(\boldsymbol{\theta}^{(l)}\big)\Big), \qquad \alpha_l = e^{\rho_l} > 0, \qquad l = 0, \dots, \tag{6}$$

with $L = 6$ layers, initialization $\boldsymbol{\theta}^{(0)} = \mathbf{0}$ (identical to the AO bound of Section I-F2, so the two are never separated by initialization luck), and $\mathrm{wrap}(\boldsymbol{\theta}) = \boldsymbol{\theta} - 2\pi\lfloor \boldsymbol{\theta}/2\pi \rfloor$, the projection back onto $[0, 2\pi)^N$ under which the rate (4) is exactly periodic. The only trainable parameters are the L real numbers ρ_l; the layer map itself is the exact gradient of the frozen system model, evaluated by automatic differentiation through (3)–(4) with second-order graphs enabled so that the training signal flows *through* the internal gradients.

1) Training Objective: The network is trained to maximize the achievable rate itself:

$$\mathcal{L}_{\mathrm{rate}} \;=\; -\,\frac{1}{|\mathcal{B}|} \sum_{\boldsymbol{h} \in \mathcal{B}} R\big(\boldsymbol{\theta}^{(L)}(\boldsymbol{h})\big), \tag{7}$$

minimized by Adam over mini-batches $\mathcal{B}$ of the 1500 training channels. This is the central design correction of this work: our earlier experiments trained an unfolding network with

mean-squared error against wrapped phase angles, which (i) optimizes a surrogate disconnected from the rate and (ii) is discontinuous at the $\pm\pi$ branch cut, where identical phases incur maximal loss. We retain both defective objectives as ablations: a smooth circular surrogate $\mathcal{L}_{\text{circ}} = \text{mean}\big(1 - \cos\big(\text{wrap}(\boldsymbol{\theta}^{(L)} - \boldsymbol{\theta}_{\text{AO}})\big)\big)$ against the AO reference phases, and the original wrapped-angle MSE $\mathcal{L}_{\text{MSE}} = \text{mean}\big(\|\,\text{wrap}\,\boldsymbol{\theta}^{(L)} - \text{wrap}\,\boldsymbol{\theta}_{\text{AO}}\|^2\big)$, reproducing its deficit in Section I-F.

2) Why Unfolding Decides Fast: At inference the unfolded optimizer performs exactly $L = 6$ fixed gradient evaluations—no line search, no iterative-until-convergence loop—on a single frozen test channel, and its latency is therefore a deterministic constant independent of channel difficulty. The Riemannian AO bound of Section I-F2, by contrast, iterates with Armijo line searches until a first-order tolerance is met. The measured per-decision latencies and the resulting quality–latency trade are reported in Section I-F; the claim we defend is the *relative* one (unfolding is orders of magnitude faster than the bound it approximates), with the absolute milliseconds table reported as measured on the training dtype.

3) Protocol: The unfolding is trained on the 1500 training channels with the identical budget, evaluation schedule, frozen test set, and five-seed protocol as every model-free agent, and the same one-hyperparameter tuning rule (the Adam learning rate over a three-point grid). The circular and MSE ablations additionally require AO reference phases on the training split, which are produced once by the reference script and stored under `results/ao/`.

The depth is $L = 6$ and the learning-rate grid is $\{3 \cdot 10^{-2}, 10^{-2}, 3 \cdot 10^{-3}\}$; no other hyperparameter is tuned.

E. Model-Driven Reference: Riemannian AO and Deep Unfolding

1) Riemannian bound: We treat (**??**) as maximization of $f(\boldsymbol{\theta})$ over the torus $\mathbb{T}^N$. Writing $\mathbf{z} = e^{j\boldsymbol{\theta}}$ and $\mathbf{g} = \nabla_{\boldsymbol{\theta}} f \in \mathbb{R}^N$, the Riemannian gradient is the Riesz representative

$$\mathbf{r}(\mathbf{z}) \;=\; j\,\mathbf{z} \odot \mathbf{g}, \qquad \langle \mathbf{r}, \mathbf{t} \rangle = df(\mathbf{t}) \;\; \forall\, \mathbf{t} = j\,\mathbf{z} \odot \mathbf{w},\; \mathbf{w} \in \mathbb{R}^N, \tag{8}$$

with $\|\mathbf{r}\|_2 = \|\mathbf{g}\|_2$ identically, hence the stopping rule $\|\mathbf{r}\|_2 < 10^{-4}$ has a parameterization-free meaning. Conjugate gradient with Fletcher–Reeves $\beta_t = \|\mathbf{r}_t\|^2 / \|\mathbf{r}_{t-1}\|^2$ and drift correction through the tangential projector $\mathcal{P}_T(\mathbf{z})(\mathbf{v}) = \mathbf{v} - \text{Re}(\mathbf{v} \odot \mathbf{z}^*) \odot \mathbf{z}$ feeds the first-order-consistent retraction $\mathcal{R}_{\mathbf{z}}(\gamma \mathbf{d}) = (\mathbf{z} + \gamma \mathbf{d})/|\mathbf{z} + \gamma \mathbf{d}|$ with Armijo backtracking ($c = 10^{-4}$, $\gamma_0 = 1$, halvable to 2^{-20}, restarts every ten accepted steps and whenever ascent pairing degrades). We run at least 200 iterations per channel. This AO serves two roles: the *optimization bound* for the study (the "AO $\geq 1.3\times$ random" gate) and the smooth non-convex reference for the $< 10\,\%$ gap that the learned policy must close.

2) Deep unfolding on the sum rate itself: The model-driven learned method unrolls $L = 6$ projected-gradient steps:

$$\boldsymbol{\theta}_{\ell+1} = \text{wrap}\Big(\boldsymbol{\theta}_\ell + e^{\alpha_\ell}\, \nabla_{\boldsymbol{\theta}_\ell} R\Big), \qquad \ell = 0, \ldots, L-1, \tag{9}$$

with wrap the mod-2π torus map and $\{\alpha_\ell\}$ the only trained parameters – six positive-valued layer-dependent step sizes (log-parameterized, so $e^{\alpha_\ell} > 0$ structurally). Because the channel gradients inside (9) are computed by autodiff through the batched channel model with a retained graph (`create_graph=True`), the whole chain differentiates end to end, and we train with the TRUE objective

$$\mathcal{L}_{\text{rate}} \;=\; -\,\mathbb{E}_{\text{ch}}\big[R(\boldsymbol{\theta}_L)\big]\,, \tag{10}$$

never an angle-space surrogate. Two supervised ablations

$$\mathcal{L}_{\text{circ}} = \mathbb{E}\big[1 - \cos(\boldsymbol{\theta}_L - \boldsymbol{\theta}^{\text{AO}})\big], \qquad \mathcal{L}_{\text{mse}} = \mathbb{E}\big[\|\text{wrap}(\boldsymbol{\theta}_L) - \text{wrap}(\boldsymbol{\theta}^{\text{AO}})\|_2^2$$

regress onto our own AO (teacher computed once on the training split). $\mathcal{L}_{\text{circ}}$ is wrap-invariant; $\mathcal{L}_{\text{mse}}$ is not — it charges a $\sim (2\pi)^2$ penalty to physically identical phases at the $0/2\pi$ boundary, exactly the failure mode destroyed our earlier study's unfolding, and we keep it only as a labeled ablation (§**??**, ablation table).

Algorithmic honesty preamble (kept short here; full statement in §**??**): each episode observes one channel and commits one phase vector – a contextual bandit, not an MDP. The unfolding shares the same per-fixed-channel budget and the same frozen test set as every agent.

3) Inference-time behavior: At test time the policicy evaluates (9) L times; end-to-end decision latency is measured and reported next to the AO wall-clock (mean/p95 per-channel), evidencing the sub-millisecond regime against the seconds-scale bound (§**??**, latency table).

F. The Riemannian Bound and the Deep Unfolding

1) Gradient Verification Before Any Result: Following the failure-analysis discipline of Section **??**, the very first experimental artifact of this work is not a rate table but a gradient audit: the autograd Euclidean gradient of $R(\boldsymbol{z})$ is compared against central finite differences along random complex directions (agreement to 10^{-5} relative or better), the Riemannian gradient is verified tangent to the torus ($\big|\text{Re}(\nabla^R R \circ \boldsymbol{z}^*)\big|_{\max} < 10^{-8}$), and the retraction is verified to return unit-modulus iterates to 10^{-12}. Every optimization result below inherits a machine-checked gradient.

2) The AO Bound (Gate 2): Riemannian conjugate gradient on the torus $\mathcal{M} = \{\boldsymbol{z} : |z_i| = 1\}$ maximizes (4) with Fletcher–Reeves directions transported by the tangent projection $P_{\boldsymbol{z}}(\boldsymbol{v}) = \boldsymbol{v} - \text{Re}(\boldsymbol{v} \circ \boldsymbol{z}^*) \circ \boldsymbol{z}$, retraction $\boldsymbol{z} \leftarrow (\boldsymbol{z} + \gamma \boldsymbol{d}) \oslash |\boldsymbol{z} + \gamma \boldsymbol{d}|$, Armijo backtracking ($c_1 = 10^{-4}$, $\gamma_0 = 1$, $\rho = 1/2$), and the stopping rule $\|\nabla^R R\|_F < 10^{-4}$ or 200 iterations, initialized at $\boldsymbol{z} = \mathbf{1}$.

On the frozen 500-channel test set the bound achieves $XX.X \pm X.X$ bps/Hz against $XX.X$ bps/Hz for random phases—a mean ratio of $XX.XX$ (criterion: ≥ 1.3), with AO beating the random-phase mean on $XX.X\%$ of channels individually. The optimizer terminates in a median of XXX iterations ($XX.X\%$ of channels meeting the first-order tolerance; the remainder terminate at a numerical local maximum with the Armijo search exhausted, both reported per channel) and takes $X.XXX$ s per channel—the "seconds-scale" reference against which learned policies are timed. A plain projected-gradient twin without the CG momentum is

included as a reviewer control and never outperforms the bound.

Fig. **??** shows the per-iteration sum-rate curve (mean and $\pm$SD band over the test set), demonstrating monotone ascent and the convergence cap at 200 iterations.

3) Unfolding Versus the Bound: 5MSX.XX The trained unfolding reaches $XX.X \pm X.X$ bps/Hz on the frozen test set—within XX.X% of the AO bound (criterion: $\leq 10\%$)—while deciding in a median of X.XX ms per channel ($p95$: 995MS ms), a $XXX\times$ speed-up over the bound, on identical channels, budgets, and seeds. The loss-ablation table confirms the design correction quantitatively: $\mathcal{L}_{\text{rate}}$ attains XX.X bps/Hz, the smooth circular surrogate XX.X bps/Hz, and the legacy wrapped-angle MSE XX.X bps/Hz—the regime that silently produced a policy indistinguishable from random phases in the earlier study.

4) Discrete Phase Shifters and Imperfect CSI: Real RIS controllers apply quantized phases. Re-mapping the AO, unfolding, and random solutions onto uniform B-bit codebooks ($B \in \{2, 3, 4\}$, $\Delta = 2\pi/2^B$) costs X.X/X.X/X.X dB at $2/3/4$ bits respectively (Table **??**), in line with the literature's 2-bit practicality sweet spot.

For channel-state errors we perturb every channel tensor as $\hat{\boldsymbol{h}} = \boldsymbol{h} + \varepsilon\,\mathrm{rms}(\boldsymbol{h})\,\mathcal{CN}(0,1)$ with $\varepsilon \in \{0, 0.01, 0.02, 0.05, 0.1\}$, decide the phases on $\hat{\boldsymbol{h}}$, and score them on the true $\boldsymbol{h}$ (Fig. **??**). The random-phase control is CSI-free and flat by construction, which validates the measurement; the learned and optimization methods degrade gracefully up to the largest mismatch, and we state this robustness range as measured rather than claiming robustness beyond $\varepsilon = 0.1$.

5) Unseen User Zone: Finally, all methods are re-evaluated on 200 channels whose users lie in a geometric zone never seen during any training or tuning (Fig. **??**). The AO bound and the unfolding degrade by X.X and X.X respectively—both optimization-based methods transfer essentially without loss because their inductive bias is the physics of (4) itself, not the training distribution. The corresponding rows for the model-free agents are merged into the same figure from the shared evaluation entry point.

G. Optimization Bound and the Model-Driven Layer

1) The AO bound is real: On the frozen 500-channel test set, Riemannian AO achieves a mean sum rate of **63.040** $\pm$**0.644** bps/Hz against **1.445**$\times$ the random-phase baseline (paired evaluation, same drawn channels, same budget). The bound gate demanded $\geq 1.3\times$; observed: **1.445** (verdict: **PASS**). Convergence metadata: mean **200** accepted iterations per channel (max-iteration budget 200; full per-channel records in the artifact `results/ao/gate2_report.json`). The plain projected-gradient twinâ€" identical machinery without conjugacyâ€"lands at **1.433**$\times$, isolating the contribution of conjugate search (Fig. **??**).

2) Unfolding closes the gap cheaply: Trained end-to-end on the rate (10), the $L = 6$ unrolled stepper reaches **94.7**% of the AO rate on the frozen test set (trigger: $\leq 10\,\%$), while its decision latency is **6.732** ms per channel versus **475** ms per channel for AO (mean; p95 **TBD** ms). Loss ablations (Table **??**) reproduce the earlier failure mechanism exactly: training on the rate dominates; the circular loss lags slightly; the MSE-on-wrapped-angles objective collapses â€" it optimizes an objective that chapters phases across the $0/2\pi$ boundary (Fig. **??**).

3) Discretized and imperfect world: Post-hoc quantization of the CONTINUOUS solutions onto uniform 2^B per-element codebooks gives the model-driven side of the bits claim (Table **??**; Shreyansh's trained-codebook DQN numbers join in Â§**??**). At $B = 3$ the AO-vs-AO-quantized gap is **1.402** dB-equivalent; unfolding shows the same ordering. Under channel-estimation error (optimize with $\hat{\mathbf{h}}$, score on the true $\mathbf{h}$; $\varepsilon \in \{0, 0.01, 0.02, 0.05, 0.1\}$, relative-Frobenius model, Â§I-E), AO degrades from **63.040** (no CSI error) to **62.861** ($\varepsilon = 0.1$), while random phases move **43.640** $\rightarrow$ **43.640** (Fig. **??**) â€" honest discussion in Â§**??**.

4) Generalization: On the `gen_channels` split (200 user positions in an UNSEEN zone, never used for any tuning), the same ordering persists: AO **56.560**, PG **TBD**, unfolding **53.422**, random **37.113**(Fig. **??**); DRL-agent columns merge in from Section **??** through the shared `evaluate.py` entry.